11. REVISION : ORIGIN OF BODY ASYMMETRY

DURATION : 03:11

STUDY SHEET

COURSE SUMMARY

This video explores the origins of bodily asymmetry, examining the movement of the notochordal process and its role in the formation of Hensen's node. Through this process, researchers discover the importance of microcilia, linked to laterality disorders and influencing the position of organs such as the heart and liver. The animation of a cascade of specific molecules and proteins is highlighted, emphasising their impact on the body's asymmetrical organisation. In addition to a theoretical aspect, this lesson offers valuable insights for clinical application, strengthening the understanding of developmental mechanisms in biodynamic embryology and osteopathy.

ORIGIN OF BODY ASYMMETRY

The origin of this **asymmetry** occurs during the movement of the **notochordal process**. This movement generates a knot, called **Hensen's node**. This node is essential in the formation of the notochordal axis.

Inside this structure, **microcilia** are found. Thanks to **Kartagener's syndrome**, Japanese researchers discovered that this system was linked to **laterality** disorders. For example, some people develop their heart on the right instead of the left. They first observed these small cilia in the region of Hensen's node.

These microcilia, present in various places in the body such as **spermatozoa**, the **uterus**, the **fallopian tubes**, and the **bronchi**, perform a rotational movement rather than a back-and-forth movement. These cilia, inclined at 60 degrees, create a movement that concentrates small molecules on one side, pushed towards the left side of the embryo.

When these small **vesicles** burst, they release a new **subtle snow of premorphogenetic molecules**, leading to internal asymmetry. Consequently, organs like the heart develop more on the left, while other organs, such as the liver, also contribute to this asymmetry.

Although the body appears symmetrical externally, a great asymmetry exists internally, whether at the **cerebral**, **ocular**, or whole-body level. This asymmetry extends between the **sacrum** and the **base of the skull**, representing the organisation of embryonic laterality.

A cascade of these small molecules releases **specific proteins**, known as **sonotic HETCH**, as well as specific genes containing **zinc**. These elements activate one side of the **corrupted** and **hypercalcium** system. This calcium activity plays a crucial role in cell development, promoting faster development on one side compared to the other.

FIG. — Autostereogram